

Transit Gateway

A Regional Hub for Scalable, Transitive Network Connectivity

Interview Preparation Guide

Problem & motivation - attachments & route tables - transitive routing - packet flow - failure modes - scaling - security - observability & automation - cost - troubleshooting - real-world patterns

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Executive Summary

A Transit Gateway (TGW) is a regional network hub that interconnects VPCs, VPN connections, and Direct Connect gateways through a single attachment per network. It replaces the operationally explosive full-mesh of VPC peering (which is non-transitive and $O(n^2)$) with a hub-and-spoke model that supports transitive routing and segmentation via TGW route tables. TGW is the backbone of large multi-account, multi-VPC, hybrid architectures. This guide covers internals, packet flow, failure modes, scaling, security, observability, automation, cost, and troubleshooting.

1. The Problem It Solves

VPC peering is simple for a few VPCs but does not scale: it is non-transitive (A-B and B-C does not give A-C) and the number of connections grows quadratically. Connecting dozens of VPCs and on-prem links by peering becomes an unmanageable mesh. A Transit Gateway centralizes this into one hub.

Core needs the TGW addresses

- **Transitive routing:** Let any attached network reach any other (when policy allows) through the hub.
- **Scale:** $O(n)$ attachments instead of $O(n^2)$ peerings.
- **Segmentation:** Use multiple TGW route tables to control who reaches whom.
- **Hybrid consolidation:** Terminate many VPN/DX connections at one hub.

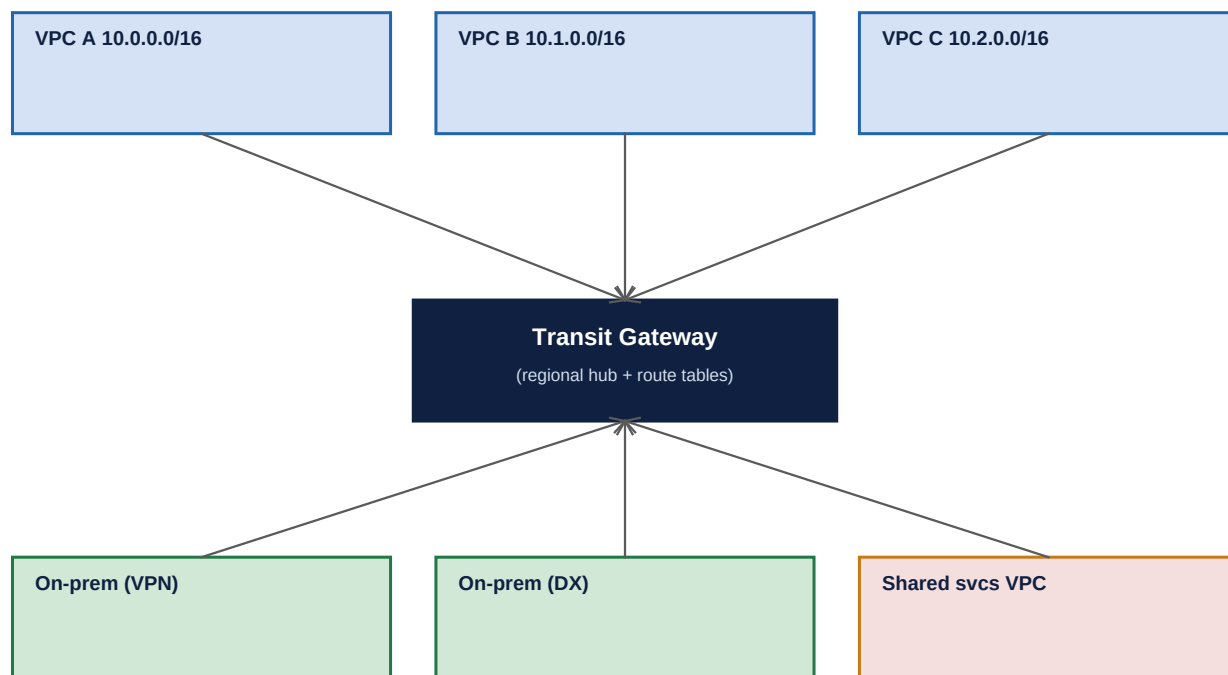
2. How It Works Internally

A TGW is a regional, managed, horizontally scaled router. Networks connect via attachments (VPC, VPN, Direct Connect gateway, or peering to another TGW). The TGW maintains one or more TGW route tables; each attachment is associated with a route table and can propagate its routes into others. This association/propagation model is how you implement segmentation (e.g., isolated spokes that all reach shared services but not each other).

- VPC attachments place TGW ENIs in chosen subnets (ideally one per AZ) for the data path.
- Subnet route tables in each VPC send the relevant prefixes to the TGW attachment.
- TGW route tables then decide the next attachment (transitive routing).
- Inter-region peering connects TGWs across regions for a global backbone.

Reference Diagram - Transit Gateway hub-and-spoke

A Transit Gateway (TGW) is a regional hub that connects many VPCs and on-prem networks through a single attachment per network, replacing an unmanageable mesh of peering connections. TGW route tables control which attachments can reach which.



Each VPC/VPN/DX attaches ONCE to the TGW. TGW route tables segment reachability (e.g., spokes reach shared services but not each other). This replaces $O(n^2)$ peering with $O(n)$ attachments and supports transitive routing.

3. Traffic Flow, Packet Flow & Routing Decisions

Packet from VPC A to VPC C via TGW

- Instance in VPC A sends to a VPC C address; SG egress evaluated.
- VPC A subnet route table: VPC C CIDR -> tgw attachment.
- Packet reaches the TGW; the attachment's associated TGW route table is consulted.
- TGW route table: VPC C CIDR -> VPC C attachment; packet forwarded.
- VPC C subnet route table must have a return route for VPC A via its TGW attachment.

Two-layer routing

TGW connectivity requires BOTH the VPC subnet route tables (to/from the TGW) AND the TGW route tables (between attachments) to be correct - in both directions. Missing either layer breaks the path.

4. Advantages & Disadvantages

Advantages

- Transitive, hub-and-spoke routing at scale.
- Segmentation via multiple route tables.
- Single place to terminate hybrid (VPN/DX) connectivity.
- Managed, HA, and inter-region capable.

Disadvantages / costs

- Per-attachment hourly charge plus per-GB data processing.
- Adds a routing hop (latency) versus direct peering.
- Central hub is a logical concentration point - design for blast radius.
- Two-layer routing increases configuration complexity.

5. Common Failure Modes

Failure	Symptom	Typical root cause
Missing VPC route	No reachability	Subnet table lacks a route to the TGW attachment.
TGW route gap	Some pairs unreachable	TGW route table missing the destination attachment.
No propagation/assoc	Segmentation wrong	Attachment not associated/propagated as intended.
CIDR overlap	Ambiguous routing	Two attached VPCs share a range.
Single-AZ attachment	Partial outage	TGW attachment ENI in too few AZs.
Asymmetric path	One-way	Return-side VPC/TGW routes missing.

6. Scaling Considerations

- **Attachments not mesh:** Add VPCs as attachments; avoid peering sprawl.
- **Route-table design:** Use separate TGW route tables for prod/dev/shared segmentation.
- **Per-AZ attachments:** Attach in every AZ you use for resilience and to avoid cross-AZ hops.
- **Non-overlapping CIDRs:** Mandatory across all attached networks; use IPAM.
- **Bandwidth:** TGW scales horizontally; per-attachment/flow limits apply at extreme scale.

7. Security Implications

- Use route-table segmentation as a security boundary (e.g., isolate spokes).
- Centralize egress/inspection by routing spokes through a firewall VPC via TGW.
- Remember TGW does not filter packets - rely on SGs/NACLs/firewall.
- Control attachment creation with IAM + RAM sharing in multi-account setups.

8. Observability, Reliability & Automation

Observability

- TGW Flow Logs for inter-attachment traffic.
- CloudWatch metrics: bytes/packets in/out per attachment, drops.
- Network Manager to visualize the global topology.

Reliability

- TGW is managed and HA within a region; attach across AZs.
- Use inter-region peering for cross-region resilience.

Automation

- Manage attachments, route tables, associations, and propagations in IaC.
- Share TGW via Resource Access Manager across accounts programmatically.

9. Cost Considerations

Cost driver	Why it adds up	Mitigation
Attachment-hours	Each attachment bills per hour.	Consolidate; remove unused attachments.
Data processing	Per-GB through the TGW.	Keep chatty workloads in one VPC; use endpoints for AWS services.
Cross-AZ/region	Extra transfer for non-local paths.	Per-AZ attachments; mind inter-region peering costs.

10. Troubleshooting Approach

For 'VPC A cannot reach VPC C via TGW':

- 1. VPC A subnet route: destination CIDR -> TGW attachment?
- 2. TGW route table for A's attachment: route to C's attachment?
- 3. Association/propagation set so the route exists where needed?
- 4. Return path: C's subnet + TGW routes back to A?
- 5. No CIDR overlap; confirm with Network Manager / Reachability Analyzer.

Mnemonic

'VPC route -> TGW route -> Assoc/Prop -> Return.' Always verify BOTH the VPC layer and the TGW layer, in BOTH directions.

11. Real-World Examples & Patterns

- **Hub-and-spoke:** Many app VPCs attach to one TGW; a shared-services VPC is reachable by all.
- **Segmented spokes:** Separate TGW route tables so spokes reach shared services but not each other.
- **Centralized egress:** Spokes route 0/0 to TGW -> egress VPC with NAT + firewall.
- **Global backbone:** Inter-region TGW peering links regional hubs; on-prem via DX gateway.

12. Senior-Level Trade-offs & Talking Points

- TGW (scalable, transitive, segmented) vs peering (lower latency/cost, non-transitive) - choose by VPC count and topology.
- Central hub simplicity vs blast-radius/bottleneck concentration.
- Per-AZ attachments (resilient, no cross-AZ fees) vs fewer attachments (cheaper, riskier).
- Centralized inspection via TGW vs distributed controls.

Whiteboard checklist

(1) Draw TGW hub with VPC/VPN/DX spokes. (2) Show single attachment each. (3) Two-layer routing (VPC + TGW tables). (4) Segment with route tables. (5) Discuss transitivity, cost (attachment + per-GB), and CIDR planning.

13. Quick Glossary

Term	Meaning
Transit Gateway	Regional hub interconnecting VPCs and on-prem.
Attachment	A network's single connection to the TGW (VPC/VPN/DX/peer).
TGW route table	Controls which attachment reaches which (segmentation).
Association	Binds an attachment to a TGW route table.
Propagation	Injects an attachment's routes into a TGW route table.
Transitive routing	A can reach C through the hub via B-less topology.
Inter-region peering	Connecting TGWs across regions.